

RAJcinema

A Full-Stack Movie Recommendation System

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Tech Stack: React (Vite) + Django REST Framework + SQLite3

Purpose: Professional portfolio showcase & introduction doc

Status: Completed and verified running

Project Architecture & Key Features

RAJcinema is a highly-optimized movie cataloging and recommendation portal built on a modern decoupled architecture. By segregating the codebase into an independent frontend single-page application (SPA) and a solid backend REST API, the system achieves maximum modularity, speed, and scalability.

Core Features

- **Dynamic Movie Grid:** Visual grid displaying fully-responsive movie cards with live TMDB posters.
- **Smart Search & Filtering:** Instant client-side search by Title, Director, or Genre, and sorting by rating/year.
- **Interactive Details Modal:** Detailed popups with synopses, vibes, genre chips, and live trailer simulations.
- **Watchlist Sync:** Personalized user watchlist that stores data persistently via browser LocalStorage.
- **Django Admin Control:** Complete administrative control to manage, edit, and add movie records directly from the database.

System Flow Architecture



Database Schema & Model Design

The project leverages an SQLite3 database backend configured via Django's Object-Relational Mapper (ORM). The primary database table is **movies_movie**, which holds structured data for the recommendation feed.

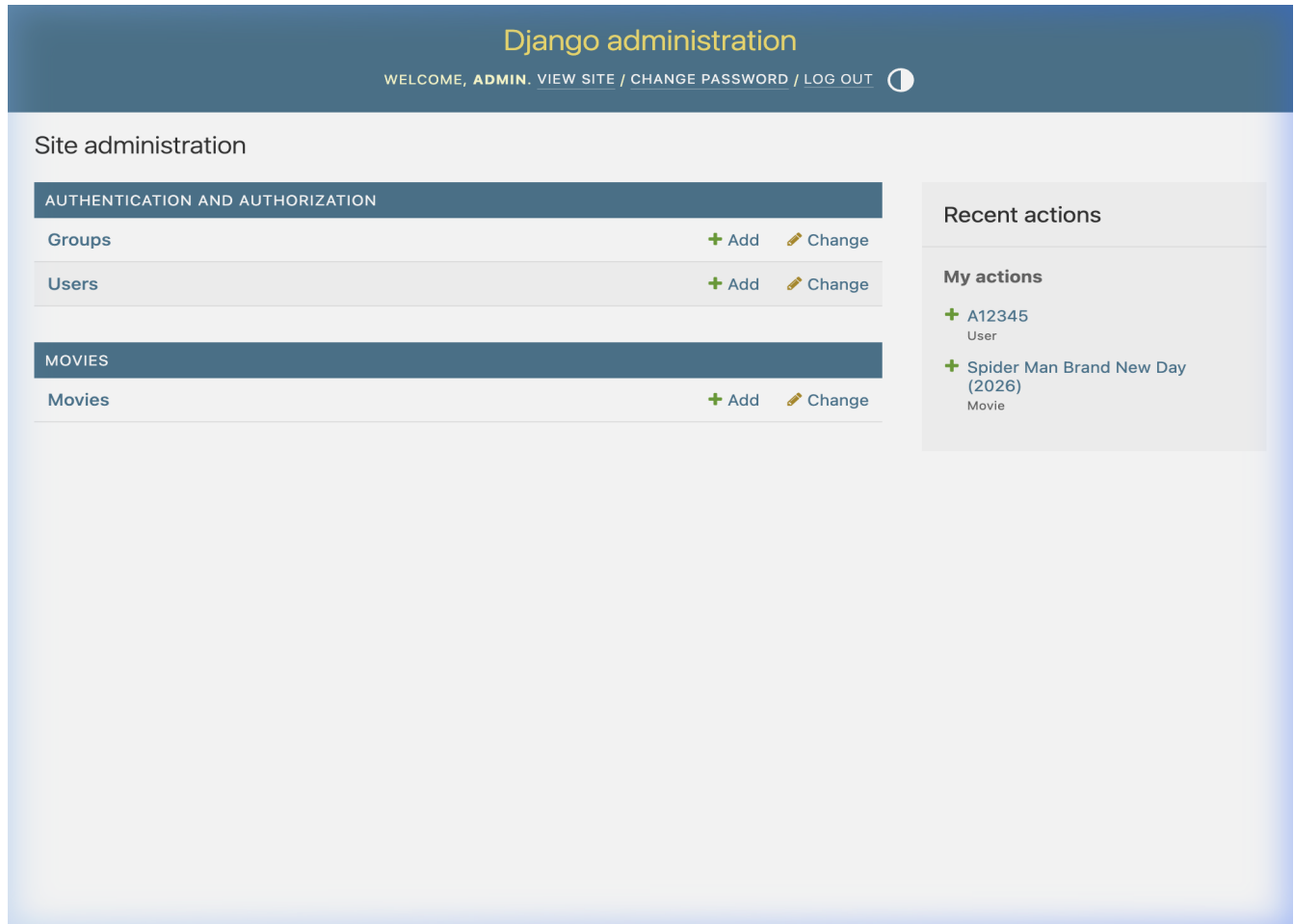
Django Data Model Definition (movies/models.py):

```
class Movie(models.Model):
    title = models.CharField(max_length=255)
    year = models.IntegerField()
    rating = models.FloatField()
    duration = models.CharField(max_length=50)
    description = models.TextField()
    director = models.CharField(max_length=255)
    poster = models.URLField(max_length=1000)
    genres = models.JSONField(default=list) # Stores genres e.g., ["Comedy", "Drama"]
    vibe = models.CharField(max_length=100)
    intensity = models.IntegerField() # Intensity rating out of 5
    era = models.CharField(max_length=50)
```

Seed Dataset: The database is initialized with 30 curated high-quality movie entries. These include major Bollywood classics (e.g. *3 Idiots*, *Taare Zameen Par*, *12th Fail*), animated blockbusters, and Hollywood favorites, mapped to release eras (classic, modern) and vibes for precise filtering.

Backend - Django Admin Dashboard

The Django Admin Dashboard allows simple site administration. It is fully authenticated, allowing authorized administrators to oversee, add, edit, or delete movies. Groups and users management is also integrated using Django's default authorization system.



Backend - Movies Model Management

The Django Admin Movies view displaying the list of all seeded movie objects. From this panel, administrators can click on any individual movie to inspect its properties or perform updates.

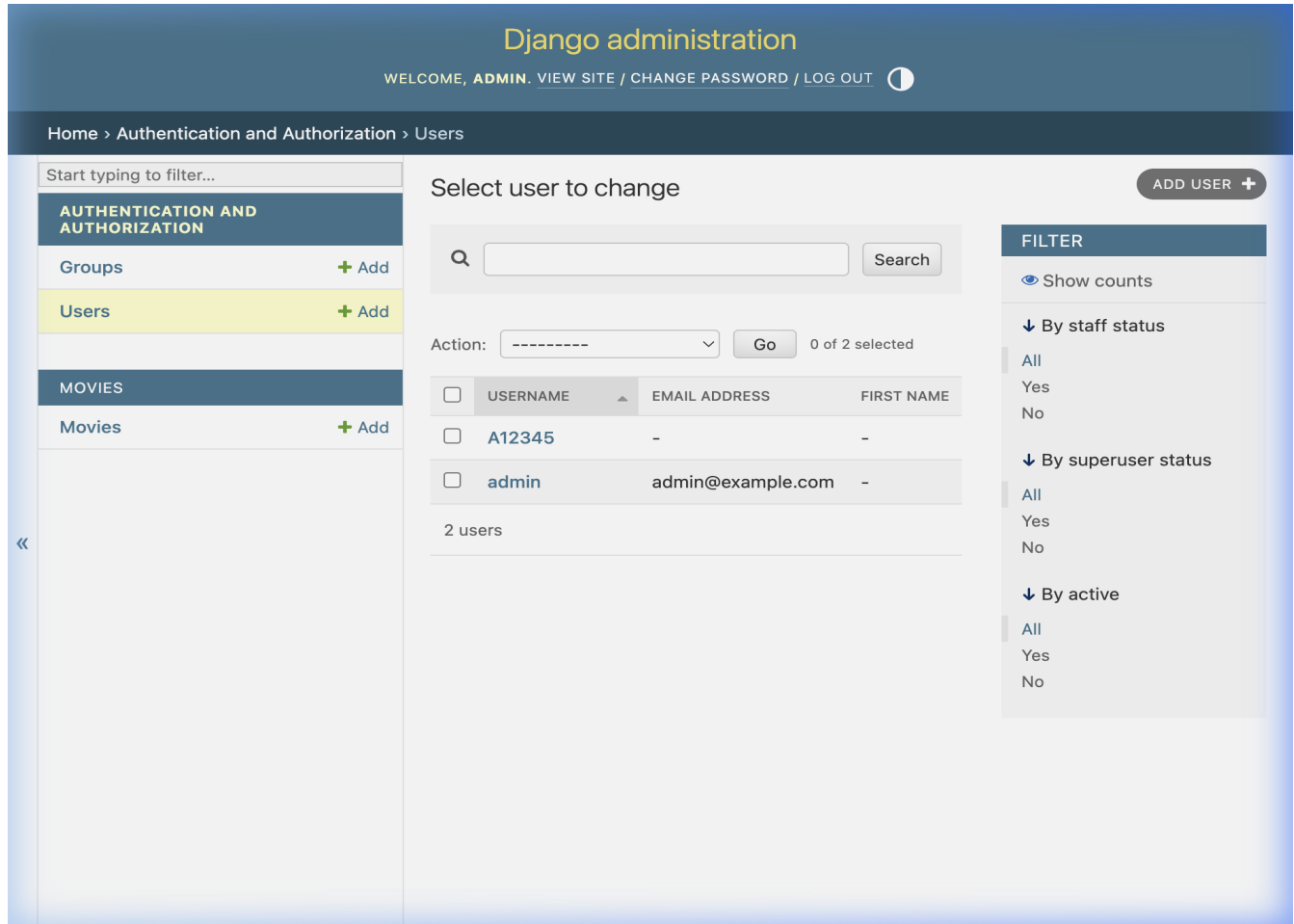
The screenshot displays the Django Admin interface for the 'Movies' model. The left sidebar shows the navigation menu with 'Home > Movies > Movies' and options to 'Add' new 'Groups', 'Users', and 'Movies'. The main content area is titled 'Select movie to change' and features a search bar and a table of 31 movies. The table columns are 'TITLE', 'YEAR', 'RATING', 'DIRECTOR', and 'VIBE'. The right sidebar contains a 'FILTER' section with 'By year' and 'By Vibe' dropdowns. The 'By Vibe' filter is expanded, showing a list of 15 movie genres.

TITLE	YEAR	RATING	DIRECTOR	VIBE
Spider-Man: Brand New Day	2025	4.0	Daniel Crittan	High-energy, emotional, youthful, urban, heroic
How to Train Your Dragon	2010	8.1	Dean DeBlois, Chris Sanders	Adventurous
Inside Out	2015	8.1	Pete Docter, Ronnie Del Carmen	Bittersweet
Ratatouille	2007	8.1	Brad Bird, Jan Pfae	Inspirational
Spider-Man: Across the Spider-Verse	2023	8.7	Joaquim Dos Santos, Kemp Powers, Justin C. Thompson	Mind-Blowing
Up	2009	8.3	Pete Docter, Bob Peterson	Whimsical
Toy Story	1995	8.3	John Lasseter	Nostalgic
Coco	2017	8.4	Lee Unkrich, Adrian Molina	Emotional
WALL-E	2008	8.4	Andrew Stanton	Heartwarming
Spider-Man: Into the Spider-Verse	2018	8.4	Bob Persichetti, Peter Ramsey, Rodney Rothman	Stylized, Electric
The Lion King	1994	8.5	Roger Allers, Rob Minkoff	Epic
777 Charlie	2022	8.8	Kiranraj K.	Heartwarming
Drishyam	2013	8.2	Jeethu Joseph	Suspenseful
Jai Bhim	2021	8.8	T.J. Gnanavel	Intense
K.G.F: Chapter 1	2018	8.2	Prashanth Neel	Gritty
Baahubali 2: The Conclusion	2017	8.2	S.S. Rajamouli	Epic
Spider-Man: No Way Home	2021	8.2	Jon Watts	Nostalgic
Avengers: Infinity War	2018	8.4	Anthony Russo, Joe Russo	Devastating
Guardians of the Galaxy	2014	8.0	James Gunn	Fun & C
The Avengers	2012	8.0	Joss Whedon	Blockbuster Thrill
Iron Man	2008	7.9	Jon Favreau	Charismatic
Dil Chahta Hai	2001	8.1	Farhan Akhtar	Nostalgic
12th Fail	2023	8.8	Vidhu Vinod Chopra	Resilient
Sholay	1975	8.1	Ramesh Sippy	Classic Masala
Swades	2004	8.2	Ashutosh Gowariker	Soulful
Andhadhun	2018	8.2	Sriram Raghavan	Suspenseful
Bang De Basanti	2006	8.1	Rakeysh Omprakash Mehra	Rebellious
Gangs of Wasseypour	2012	8.2	Anurag Kashyap	Visceral
Dangal	2016	8.3	Nitesh Tiwari	Gritty
Taare Zameen Par	2007	8.3	Aamir Khan	Emotional
3 Idiots	2009	8.4	Rajkumar Hirani	Inspirational

31 movies

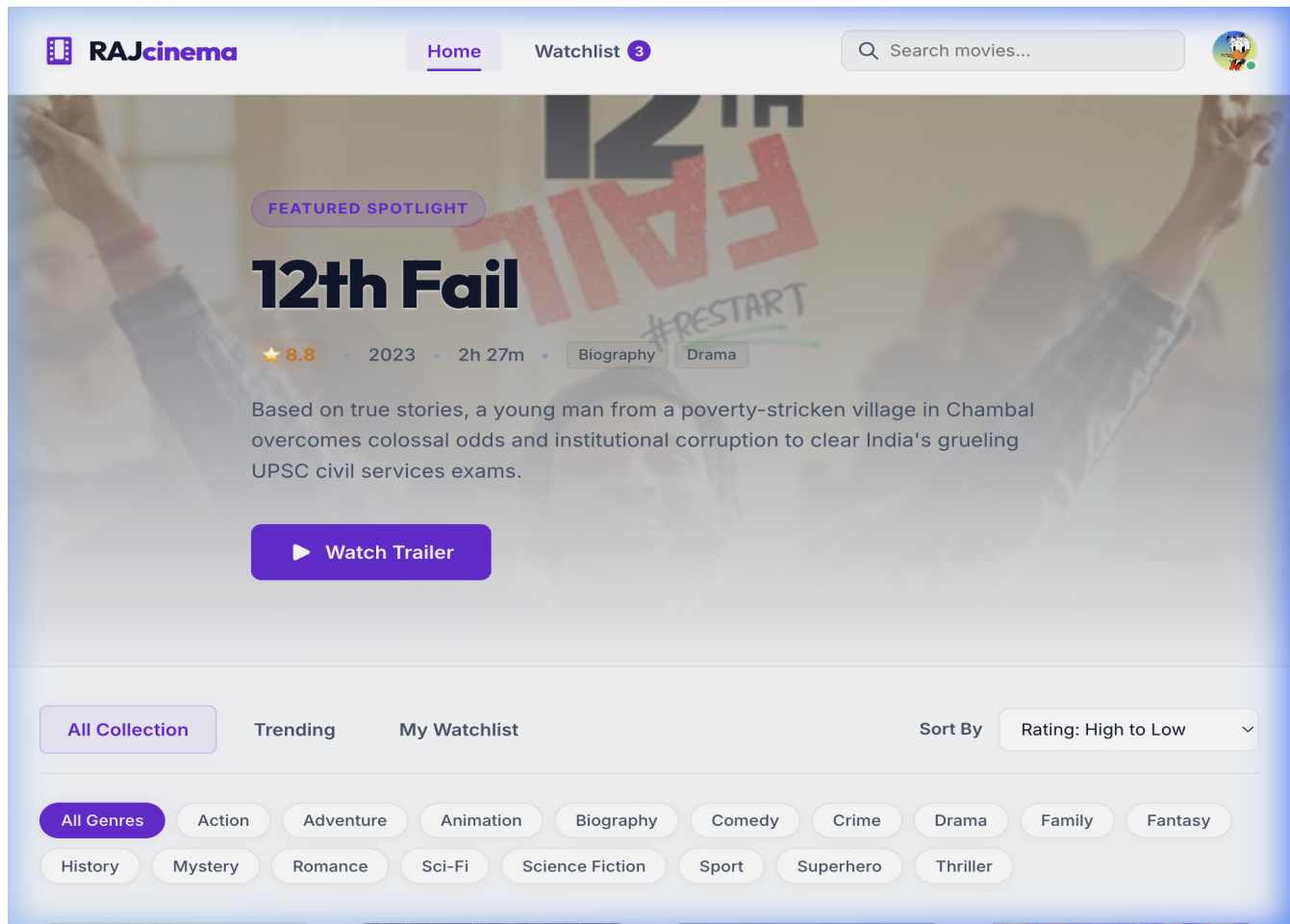
Backend - User Accounts Control

The Users list view in Django Admin. Currently, the system contains the default admin superuser account used for setup and site management.



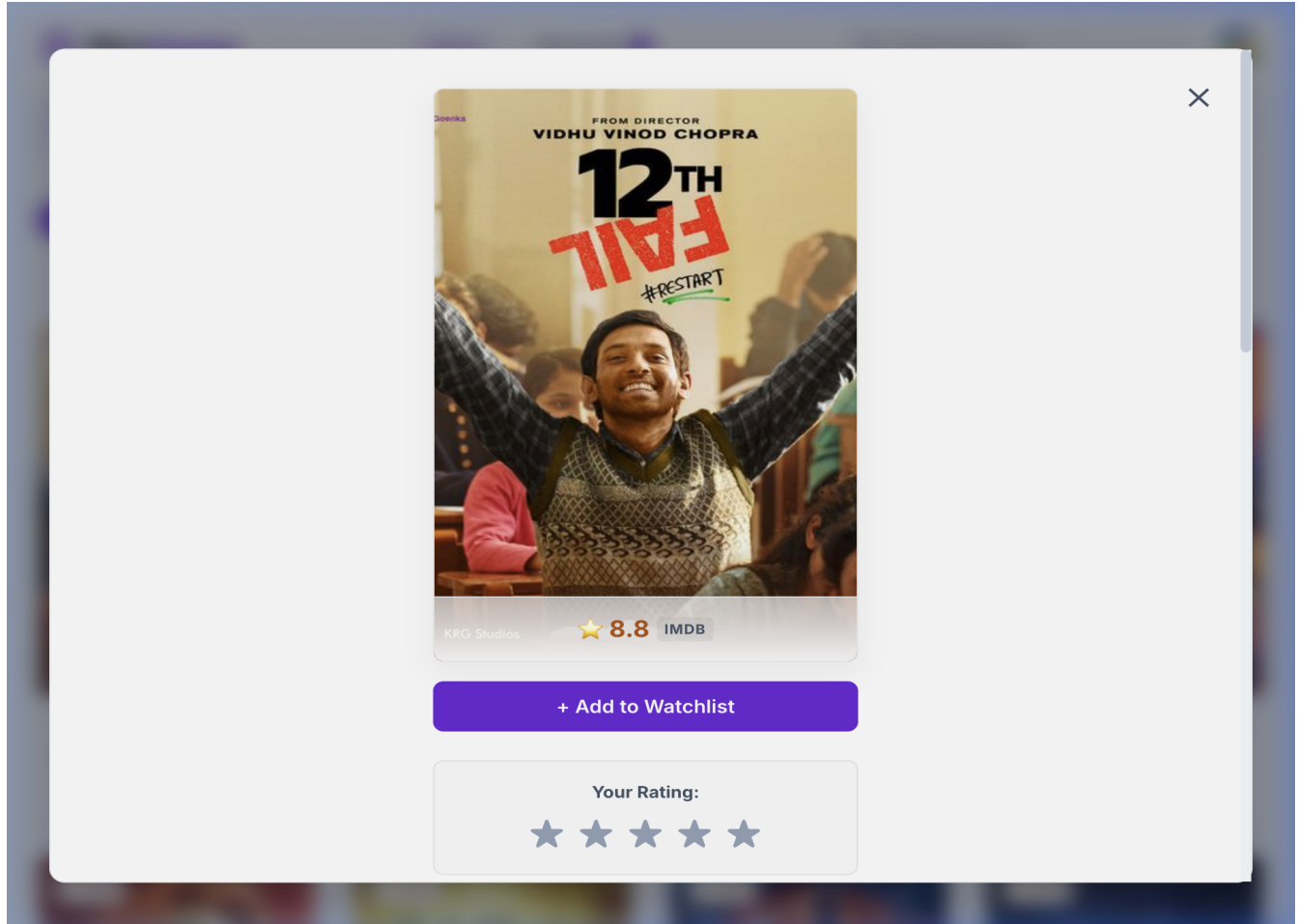
Frontend - Home Page & Spotlight

The premium landing page of RAJcinema. Built using React and a Warm Slate CSS theme, it displays the spotlight movie (12th Fail) as the Hero banner and dynamically loads movie cards using AJAX calls to the Django server. Sorting (by rating, release year, title) and genre filter chips are fully functional.



Frontend - Movie Detail Modal

When clicking on a movie card, the app loads this overlay Detail Modal. It showcases detailed movie specifications, genres list, an interactive intensity bar, and starts a simulated video trailer stream when clicked.



Frontend - Watchlist View & Sync

The Watchlist tab displays only the movies that the user has bookmarked. This feature uses custom state syncing with the browser's LocalStorage, meaning user bookmarks remain saved even after closing or refreshing the tab.

